

Exercise 1.

- bin contains stuff animals: 1 bear, 1 raccoon, 2 penguins, and 6 rabbits
- person randomly grabs 3
- Y number of different types of stuffed animals in selection

$$X_1 = 1 \text{ bear}$$

$$X_2 = 1 \text{ raccoon}$$

$$X_3 = 2 \text{ penguins}$$

$$X_4 = 6 \text{ rabbits}$$

$$Y = X_1 + X_2 + X_3 + X_4$$

$$E(Y) = E(X_1) + E(X_2) + E(X_3) + E(X_4)$$

$$E(Y) = P(X_1 = 1) + P(X_2 = 1) + P(X_3 = 1) + P(X_4 = 1)$$

$$1 - P(X_1 = 0) + 1 - P(X_2 = 0) + 1 - P(X_3 = 0) + 1 - P(X_4 = 0)$$

$$= 1 - \frac{\binom{9}{3}}{\binom{10}{3}} + 1 - \frac{\binom{9}{3}}{\binom{10}{3}} + 1 - \frac{\binom{8}{3}}{\binom{10}{3}} + 1 - \frac{\binom{4}{3}}{\binom{10}{3}}$$

$$= 4 - \frac{9!}{3!6!} \times \frac{3!7!}{10!} - \frac{9!}{3!6!} \times \frac{3!7!}{10!} - \frac{8!}{3!5!} \times \frac{3!7!}{10!} - \frac{4!}{3!} \times \frac{3!7!}{10!}$$

$$= 4 - \frac{7}{10} - \frac{7}{10} - \frac{7 \times 6}{10 \times 9} - \frac{4!7!}{10!}$$

$$= 4 - \frac{7}{10} - \frac{7}{10} - \frac{7}{15} - \frac{1}{30}$$

$$= \frac{21}{10}$$

$$= 2.1$$

Exercise 2.

- 4 pairs of shoes are lined up at random

$$E(X) = \text{number of shoes next to each other}$$

$$\Omega = 8! = 40,320$$

$$E(X_i) = \begin{cases} 1, & \text{probability of any pair } (1, 2, 3, 4) \\ 0, & \text{probability of no pair} \end{cases}$$

$$P(X = 0) = \binom{4}{0} 8! = 40,320 \rightarrow \frac{40,320}{40,320} = 1$$

$$P(X = 1) = \binom{4}{1} 7!2 = 40,320 - 17,280 = 23,040 \rightarrow \frac{17,280}{40,320} = \frac{4}{7}$$

$$P(X = 2) = \binom{4}{2} 6!2^2 = 17,280 - 3,840 = 13,440 \rightarrow \frac{13,440}{40,320} = \frac{1}{3}$$

$$P(X = 3) = \binom{4}{3} 5!2^3 = 3,840 - 384 = 3,456 \rightarrow \frac{3,456}{40,320} = \frac{3}{35}$$

$$P(X = 4) = \binom{4}{4} 4!2^4 = 384 \rightarrow \frac{384}{40,320} = \frac{1}{105}$$

$$P(\text{Prob of any pair}) = \frac{4}{7} + \frac{1}{3} + \frac{3}{35} + \frac{1}{105} = 1$$

$$P(\text{Prob of no pair}) = 1$$

$$E(X) = 0 \times 1 + 1 \times 1 = 1$$

$$P(\text{Prob of any pair}) = \frac{\binom{4}{1} 7!2}{8!} = \frac{8 \text{ times } 7!}{8!} = 1$$

Generalization:

$$P(\text{Prob of any pair}) = \frac{\binom{n}{1} (2n-1)!2}{(2n)!} = \frac{\frac{n!}{1!(n-1)!} (2n-1)!2}{(2n)!} = \frac{2n(2n-1)!}{(2n)!} = \frac{(2n)!}{(2n)!} = 1$$

$$P(\text{Prob of no pair}) = \frac{\binom{n}{0} (2n)!}{(2n)!} = \frac{(2n)!}{(2n)!} = 1$$

$$E(X) = 1 \times 0 + 1 \times 1$$

Therefore $E(X)$ always equals 1 no matter the number of shoes

Exercise 3.

$$(a) E(X) = \begin{cases} 1, & \text{if has rank } i \\ 0, & \text{if does not have rank } i \end{cases}$$

13 ranks, 4 cards each

$$X = \sum_{i=1}^{13} X_i$$

$$E(X) = E(\sum_{i=1}^{13} X_i)$$

$$P(\text{Doesn't have rank } i) = \frac{\binom{48}{10}}{\binom{52}{10}}$$

$$P(\text{has rank } i) = 1 - \frac{\binom{48}{10}}{\binom{52}{10}}$$

$$E(X) = P(X = 1) + P(X = 2) + \dots + P(X = 13)$$

$$E(X) = 13P(X = i)$$

$$E(X) = 13\left(1 - \frac{\binom{48}{10}}{\binom{52}{10}}\right)$$

$$(b) E(X) = \begin{cases} 1, & \text{if has rank } i \\ 0, & \text{if doesn't have rank } i \end{cases}$$

$$E(X) = [13 \times (\text{ranks with 4 cards}) + 1(\text{rank with 2 cards})] \times 1 + 0 \times P(\text{doesn't have rank } i, i = 1, \dots, 13, 14)$$

$$E(X) = 13\left(1 - \frac{\binom{50}{10}}{\binom{54}{10}}\right) + \left(1 - \frac{\binom{52}{10}}{\binom{54}{10}}\right)$$

Exercise 4.

$X_1 = 1\text{st toy} \rightarrow \text{comes from the 1st draw}$

$X_2 = 2\text{nd toy} \rightarrow \text{number of draw until 2nd toy, geometric}(3/4)$

$X_3 = 3\text{rd toy} \rightarrow \text{geometric}(1/2)$

$X_4 = 4\text{th toy} \rightarrow \text{geometric}(1/4)$

$$X = X_1 + X_2 + X_3 + X_4$$

$$E(X) = E(X_1) + E(X_2) + E(X_3) + E(X_4)$$

$$E(X) = 1 + \frac{1}{3/4} + \frac{1}{1/2} + \frac{1}{1/4}$$

$$E(X) = 1 + \frac{4}{3} + 2 + 4$$

$$E(X) = \frac{25}{3}$$

Exercise 5.

$$(a) P(\text{person 4 has the winning bid}) = \frac{1}{4}$$

because everyone is equally likely

$$(b) P(X_1 > X_2 > X_3) = \frac{1}{3} \times \frac{1}{2} \times 1 = \frac{1}{6}$$

$$(c) P(X_1 > X_2 > X_3 \text{ and person 4 has the winning bid}) = \frac{1}{4} \times \frac{1}{3} \times \frac{1}{2} \times 1 = \frac{1}{24}$$

$$\begin{aligned} (d) & P(X_1 > X_2 > X_3 \text{ and person 4 does not have the winning bid}) \\ &= P(X_1 > X_2 > X_3) - P(X_1 > X_2 > X_3 \text{ and person 4 has the winning bid}) \\ &= \frac{1}{6} - \frac{1}{24} = \frac{1}{8} \end{aligned}$$

$$\begin{aligned} (e) & P(\text{person 4 has the winning bid and person 1's bid is not the lowest bid}) \\ &= P(\text{person 4 has the winning bid}) - (P(X_4 > X_3 > X_2 > X_1) + P(X_4 > X_2 > X_3 > X_1)) \\ &= \frac{1}{4} - \left(\frac{1}{24} + \frac{1}{24}\right) = \frac{1}{4} - \frac{1}{12} = \frac{1}{6} \end{aligned}$$

(f) There is not information to answer

Exercise 6.

- a box starts with 2 balls: 1 black, 1 white
- $X_i = 1$ if we draw a white ball on trial i

$$\begin{aligned} (a) & P(X_2 = 1) = \frac{1}{2} \\ & P(X_4 = 1) = \frac{1}{2} \end{aligned}$$

$$(b) P(X_2 = 1 | X_4 = 1) = \frac{P(X_2=1, X_4=1)}{P(X_4=1)} = \frac{P(X_1=1, X_2=1)}{P(X_1=1)} = \frac{\frac{1}{2} \times \frac{2}{3}}{\frac{1}{2}} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{1}{3} \times 2 = \frac{2}{3}$$

$$\begin{aligned} (c) & P(X_1 = 1, X_2 = 1, X_3 = 1) = \\ & (W \times W \times W \times W) + (W \times W \times W \times B) + (B \times W \times W \times W) \\ &= \left(\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5}\right) + \left(\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{1}{5}\right) + \left(\frac{1}{2} \times \frac{1}{3} \times \frac{2}{4} \times \frac{3}{5}\right) \\ &= \frac{24}{120} + \frac{6}{120} + \frac{6}{120} \\ &= \frac{36}{120} \\ &= \frac{3}{10} \end{aligned}$$

(d) Expected value of the number of white balls in 4 trials = $E(Y)$

$$Y = X_0 + X_1 + X_2 + X_3 + X_4$$

$$E(Y) = E(X_0) + E(X_1) + E(X_2) + E(X_3) + E(X_4)$$

$$X_i = \begin{cases} 1, & \text{if its white} \\ 0, & \text{otherwise} \end{cases}$$

$$E(Y) = 0 + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)$$

$$E(Y) = 2$$

(e) $Y = X_1 + X_2 + X_3 + X_4$

$$Var(Y) = Var(X_1 + X_2 + X_3 + X_4)$$

$$Var(Y) = E[(X_1 + X_2 + X_3 + X_4 - E[X_1 + X_2 + X_3 + X_4])^2]$$

$$Var(Y) = E[((X_1 - E[X_1]) + (X_2 - E[X_2]) + (X_3 - E[X_3]) + (X_4 - E[X_4]))^2]$$

$$Var(Y) = E[(X_1 - E[X_1])^2 + 2(X_1 - E[X_1])(X_2 - E[X_2]) + (X_2 - E[X_2])^2 + (X_3 - E[X_3])^2 + (X_4 - E[X_4])^2 + 2(X_1 - E[X_1])(X_3 - E[X_3]) + 2(X_2 - E[X_2])(X_3 - E[X_3]) + 2(X_1 - E[X_1])(X_4 - E[X_4]) + 2(X_2 - E[X_2])(X_4 - E[X_4]) + 2(X_3 - E[X_3])(X_4 - E[X_4])]$$

$$Var(Y) = Var(X_1) + 2Cov(X_1, X_2) + Var(X_2) + Var(X_3) + Var(X_4) + 2Var(X_1, X_3) + 2Var(X_2, X_3) + 2Var(X_1, X_4) + 2Var(X_2, X_4) + 2Var(X_3, X_4)$$

$$Var(Y) = \frac{1}{4} + \frac{1}{6} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$$

$$Var(Y) = 1 + 1$$

$$Var(Y) = 2$$

Exercise 7.

$$Z_1, Z_2, Z_3, Z_4 \sim IID \rightarrow N(0, 1)$$

$$(a) P(Z_4 > Z_1, Z_4 > Z_2, Z_4 > Z_3) = \frac{1}{4}$$

$$(b) P(Z_4 > Z_1, Z_4 > Z_2) = P(Z_4, Z_2, Z_1, Z_3) \times 3! + P(Z_3 > Z_4 > Z_2, Z_1) \times 2!$$

$$P(Z_4 > Z_1, Z_4 > Z_2) = \frac{1}{24} \times 6 + \frac{1}{24} \times 2$$

$$P(Z_4 > Z_1, Z_4 > Z_2) = \frac{8}{24}$$

$$P(Z_4 > Z_1, Z_4 > Z_2) = \frac{1}{3}$$

$$(c) P(Z_4 > Z_1 | Z_1 = \max\{Z_1, Z_2, Z_3\}) = \frac{P(Z_4 > Z_1, Z_1 = \max\{Z_1, Z_2, Z_3\})}{P(Z_1 = \max\{Z_1, Z_2, Z_3\})} = \frac{\frac{1}{4} \times \frac{1}{3}}{\frac{1}{3}} = \frac{1/12}{1/3} = \frac{1}{12} \times 3 = \frac{1}{4}$$

Exercise 8.

$$\begin{aligned} Cov(a, X) &= E(a - E(a))E(X - E(X)) \\ &= E(a - a)E(X - E(X)) \\ &= E(0)E(X - E(X)) \\ &= 0 \times E(X - E(X)) \\ &= 0 \end{aligned}$$

$$\begin{aligned} Cov(a, X) &= E(a - E(a))E(X - E(X)) \\ &= (E(X - E(X))E(a - E(a))) \\ &= Cov(X, a) \end{aligned}$$

$$\begin{aligned} Cov(X, a) &= E(X - E(X))E(a - E(a)) \\ &= E(X - E(X))E(a - a) \\ &= E(X - E(X))E(0) \\ &= E(X - E(X)) \times 0 \\ &= 0 \end{aligned}$$

Exercise 9.

- 2 satellites measure the distance an object travels
- satellite i return the measurement $D_i = d + \epsilon_i$
- d = actual distance
- ϵ_i = random error made by satellite i
- D_1 and D_2 are independent
- we will measure the distance $= cD_1 + (1 - c)D_2 = d$

- find constant c such that $0 \leq c \leq 1$ and $Var(cD_1 + (1 - c)D_2)$ is minimized
- $Var(D_1) = \sigma_1^2$ and $Var(D_2) = \sigma_2^2$ are both finite positive

$$D_i = d + \epsilon_i$$

$$D_i = cD_1 + (1 - c)D_2 + \epsilon_i$$

$$Var(D_i) = Var(cD_1 + (1 - c)D_2 + \epsilon_i)$$

$$= Var(cD_1 + (1 - c)D_2)$$

$$= Var(cD_1) + Var((1 - c)D_2) + 2Cov(cD_1, (1 - c)D_2)$$

$$= c^2Var(D_1) + (1 - c)^2Var(D_2) + 2c(1 - c)Cov(D_1, D_2)$$

$$= c^2Var(D_1) + (1 - c)^2Var(D_2) + (2c - 2c^2)[Cov(D_1, D_2) = 0] \text{ since independent}$$

$$= c^2\sigma_1^2 + (1 - 2c + c^2)\sigma_2^2$$

$$\frac{d}{dc}(c^2\sigma_1^2 + (1 - 2c + c^2)\sigma_2^2) = 0$$

$$\frac{d}{dc}(c^2\sigma_1^2 + \sigma_2^2 - 2c\sigma_2^2 + c^2\sigma_2^2) = 0$$

$$2c\sigma_1^2 - 2\sigma_2^2 + 2c\sigma_2^2 = 0$$

$$2c\sigma_1^2 + 2c\sigma_2^2 = 2\sigma_2^2$$

$$c(\sigma_1^2 + \sigma_2^2) = \sigma_2^2$$

$$c = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

$$= Var\left(\frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}D_1, \left(1 - \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right)D_2\right)$$

$$= Var\left(\frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}D_1\right) + Var\left(\left(1 - \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right)D_2\right) + 2Cov\left(\frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}D_1, \left(1 - \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right)D_2\right)$$

$$= \left(\frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right)^2 Var(D_1) + \left(1 - \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right)^2 Var(D_2) + 2\left(\frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right)\left(1 - \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right)Cov(D_1, D_2)$$

$$= \left(\frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right)^2 \sigma_1^2 + \left(1 - \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right)^2 \sigma_2^2 + 2\left(\frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right)\left(1 - \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right) \times 0$$

$$= \frac{\sigma_2^4}{(\sigma_1^2 + \sigma_2^2)^2} \sigma_1^2 + \left(\frac{\sigma_1^2 + \sigma_2^2 - \sigma_2^2}{\sigma_1^2 + \sigma_2^2}\right)^2 \sigma_2^2$$

$$= \frac{\sigma_2^4 \sigma_1^2}{(\sigma_1^2 + \sigma_2^2)^2} + \left(\frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}\right)^2 \sigma_2^2$$

$$= \frac{\sigma_2^4 \sigma_1^2}{(\sigma_1^2 + \sigma_2^2)^2} + \frac{\sigma_1^4 \sigma_2^2}{(\sigma_1^2 + \sigma_2^2)^2}$$

$$= \frac{\sigma_2^2 \sigma_1^2 (\sigma_2^2 + \sigma_1^2)}{(\sigma_1^2 + \sigma_2^2)^2}$$

$$= \frac{\sigma_2^2 \sigma_1^2}{\sigma_1^2 + \sigma_2^2}$$

Exercise 10.

$$(a) Z_x = \frac{x - \mu_x}{\sigma_x}$$

$$E(Z_x) = E\left(\frac{X - \mu_x}{\sigma_x}\right)$$

$$= \frac{E(X) - E(\mu_x)}{\sigma_x}$$

$$= \frac{\mu_x - \mu_x}{\sigma_x}$$

$$= 0$$

$$Var(Z_x) = Var\left(\frac{X - \mu_x}{\sigma_x}\right)$$

$$= \frac{1}{\sigma_x^2} Var(X - \mu_x)$$

$$= \frac{1}{\sigma_x^2} Var(X)$$

$$= \frac{\sigma_x^2}{\sigma_x^2}$$

$$= 1$$

$$(b) Corr(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

$$Corr(X, Y) = \frac{Cov(X, Y)}{\sqrt{\sigma_x^2 \sigma_y^2}}$$

$$Corr(X, Y) = \frac{E(XY) - E(Y)E(X)}{\sigma_x \sigma_y}$$

$$Corr(X, Y) = \frac{E[(X - E(X))(Y - E(Y))]}{\sigma_x \sigma_y}$$

$$Corr(X, Y) = \frac{E[(X - \mu_x)(Y - \mu_y)]}{\sigma_x \sigma_y}$$

$$Corr(X, Y) = E\left[\frac{1}{\sigma_x \sigma_y} (X - \mu_x)(Y - \mu_y)\right]$$

$$Corr(X, Y) = E\left[\left(\frac{X - \mu_x}{\sigma_x}\right)\left(\frac{Y - \mu_y}{\sigma_y}\right)\right]$$

$$Corr(x, y) = E[Z_x Z_y]$$

$$Corr(x, y) = E[(Z_x - 0)(Z_y - 0)]$$

$$Corr(x, y) = Cov(Z_x, Z_y)$$

Exercise 11.

- independent sequence $X_1, X_2, X_3, \dots \sim N(\mu, \sigma^2)$
- for each n , the random variable $\bar{X}_n = \frac{1}{n} \sum_{k=1}^n X_k$ denotes the sample means of the first n random variables in the sequence

(a) Because of the linearity property of covariance

$$\text{Cov}(X_1 + X_2, X_1 - X_2) = \text{Cov}(X_1, X_1) - \text{Cov}(X_1, X_2) + \text{Cov}(X_2, X_1) - \text{Cov}(X_2, X_2)$$

This has the following properties:

- $\text{Cov}(X_i, X_i) = \text{Var}(X_i) = \sigma^2$
- $\text{Cov}(X_1, X_2) = \text{Cov}(X_2, X_1) = 0$, because X_1 and X_2 are independent

$$\text{Cov}(X_1 + X_2, X_1 - X_2) = \sigma^2 - 0 + 0 - \sigma^2 = 0$$

(b)

- if $1 \leq m < n$ compute $\text{Cov}(\bar{X}_m, \bar{X}_n)$

$$= \text{cov}(\bar{X}_m, \bar{X}_n)$$

$$= \text{Cov}\left(\frac{1}{m} \sum_{j=1}^m X_j, \frac{1}{n} \sum_{k=1}^m X_k + \frac{1}{n} \sum_{k=m+1}^n X_k\right)$$

$$= \frac{1}{mn} \text{Cov}\left(\sum_{j=1}^m X_j, \sum_{k=1}^m X_k + \sum_{k=m+1}^n X_k\right)$$

$$= \frac{1}{mn} \text{Cov}\left(\sum_{j=1}^m X_j, \sum_{k=1}^m X_k\right) + \frac{1}{mn} \text{Cov}\left(\sum_{j=1}^m X_j, \sum_{k=m+1}^n X_k\right)$$

$$= \frac{1}{mn} \sum_{j=1}^m \sum_{k=1}^m \text{Cov}(X_j, X_k) + \frac{1}{mn} \sum_{j=1}^m \sum_{k=m+1}^n \text{Cov}(X_j, X_k)$$

$$\text{Cov}(X_j, X_k) = \begin{cases} \sigma^2, & \text{if } j = k \\ 0, & \text{if } j \neq k \end{cases}$$

$$= \frac{1}{mn} \sum_{j=1}^m \sum_{k=1}^m \text{Cov}(X_j, X_k)$$

$$= \frac{1}{mn} \sum_{j=1}^m \sigma^2$$

$$= \frac{\sigma^2}{n}$$

$$(c) \text{Cov}(X_i - \bar{X}_n, X_j - \bar{X}_n) = \text{Cov}(X_i, X_j) - \text{Cov}(X_i, \bar{X}_n) - \text{Cov}(\bar{X}_n, X_j) + \text{Cov}(\bar{X}_n, \bar{X}_n)$$

Covariance properties:

- $Cov(X_i, X_j) = 0$ since $i \neq j$, therefore X_i and X_j are independent
- $Cov(X_i, X_n) = \frac{\sigma^2}{n}$ from part (b)
- $Cov(X_n, X_n) = Var(X_n) = \frac{\sigma^2}{n}$

$$Cov(X_i - X_n, X_j - X_n) = 0 - \frac{\sigma^2}{n} - \frac{\sigma^2}{n} + \frac{\sigma^2}{n}$$

$$Cov(X_i - X_n, X_j - X_n) = -\frac{\sigma^2}{n}$$

Exercise 12.

- $N - \text{geometric}(p)$
- on day N the stock is worth X dollars
- $X - \log - \text{normal}(\mu N, \sigma^2 N)$

$$E(X|N) = e^{\mu N + \frac{\sigma^2 N}{2}} = e^{(\mu + \frac{\sigma^2}{2})N}$$

Using the law of total expression:

$$E(X) = E(E(X|N))$$

$$E(X) = E(e^{(\mu + \frac{\sigma^2}{2})N})$$

$$E(X) = \sum_{N=1}^{\infty} (1-p)^{N-1} p e^{(\mu + \frac{\sigma^2}{2})N}$$

$$E(X) = p \sum_{N=1}^{\infty} (1-p)^{N-1} e^{(\mu + \frac{\sigma^2}{2})N}$$

$$\text{Assume } \mu + \frac{\sigma^2}{2} < -\ln(1-p)$$

$$E(X) = p e^{(\mu + \frac{\sigma^2}{2})} \sum_{N=1}^{\infty} (1-p)^{N-1} (e^{(\mu + \frac{\sigma^2}{2})})^{N-1}$$

$$E(X) = p e^{(\mu + \frac{\sigma^2}{2})} \sum_{N=1}^{\infty} ((1-p)(e^{(\mu + \frac{\sigma^2}{2})}))^{N-1}$$

$$E(X) = p e^{(\mu + \frac{\sigma^2}{2})} \times \frac{1}{1 - (1-p)e^{(\mu + \frac{\sigma^2}{2})}}$$

$$E(X) = \frac{pe^{\mu + \frac{\sigma^2}{2}}}{1 - (1-p)e^{\mu + \frac{\sigma^2}{2}}}$$

Exercise 13.

- Prof comes at a time between 9 AM - 1 PM and performs task
- duration of the task is exponentially distributed with parameter $\lambda(y) = \frac{1}{5-y}$
- y is the length of time interval between 9 AM and the time of their arrival

(a) $Y \sim \text{unif}(0, 4)$

$$E(Y) = \frac{4}{2} = 2$$

$$X|Y = y \sim \text{exp}\left(\frac{1}{5-y}\right)$$

$$E(X|Y = y) = 5 - y$$

$$E(X) = E(5 - Y) = 5 - E(Y) = 5 - 2 = 3 \text{ hours}$$

(b) $E[Y + X] = E(Y) + E(X) = 3 + 2 = 5 \text{ hours}$

Exercise 14.

$X \sim \text{geometric}\left(\frac{1}{2}\right)$

$Y = \min\{X, k\}$

$$E(Y) = E(Y|A)P(A) + E(Y|A^c)P(A^c)$$

- $A = \{X \leq k\}$
- $A^c = \{X > k\}$

$$P(A) = P(X \leq k) = 1 - (1-p)^k = 1 - \left(1 - \frac{1}{2}\right)^k = 1 - \left(\frac{1}{2}\right)^k$$

$$P(A^c) = P(X > k) = (1-p)^k = \left(1 - \frac{1}{2}\right)^k = \left(\frac{1}{2}\right)^k$$

$$P(X = x) = (1-p)^{x-1}p = \left(1 - \frac{1}{2}\right)^{x-1}\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^{x-1}\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^x$$

$$E(Y|A) = \frac{\sum_{x=1}^k xP(x=k)}{P(X \leq k)} = \frac{\sum_{x=1}^k x(\frac{1}{2})^x}{1 - (\frac{1}{2})^k}$$

$$E(Y|A^c) = P(Y|X > k) = E(k|X > k) = k$$

$$E(Y) = \frac{\sum_{x=1}^k x(1/2)^k}{1 - (1/2)^k} (1 - (1/2)^k) + k \frac{1}{2^k}$$

$$E(Y) = \sum_{x=1}^k x(\frac{1}{2})^x + \frac{k}{2^k}$$

$$E(Y) = \sum_{x=1}^k \frac{x}{2^x} + \frac{k}{2^k}$$

$$E(Y) = 2^{-k}(-k + 2^{k+1} - 2) + \frac{k}{2^k}$$

$$E(Y) = 2^{-k}(2(2^k - 1) - k) + \frac{k}{2^k}$$

$$E(Y) = 2(1 - \frac{1}{2^k}) - \frac{k}{2^k} + \frac{k}{2^k}$$

$$E(Y) = 2(1 - 2^{-k})$$

Exercise 15.

$X \sim \text{Gamma}(\alpha, \beta)$

$Y|X = x \sim \text{Gamma}(X, X)$

$$(a) E(Y) = E(E(Y|X)) = E(X^2) = \text{Var}(X) + E(X)^2 = \alpha\beta^2 + \alpha^2\beta^2$$

$$(b) \text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$\text{Cov}(X, Y) = E[E(XY|X)] - (\alpha\beta)(\alpha\beta^2 + \alpha^2\beta^2)$$

$$\text{Cov}(X, Y) = E[XE(Y|X)] - \alpha^2\beta^3 - \alpha^3\beta^3$$

$$\text{Cov}(X, Y) = E[XX^2] - \alpha^2\beta^3 - \alpha^3\beta^3$$

$$\text{Cov}(X, Y) = E[X^3] - \alpha^2\beta^3 - \alpha^3\beta^3$$

$$M(\theta) = (1 - \beta\theta)^{-\alpha}$$

$$M'(\theta) = \alpha\beta(1 - \beta\theta)^{-\alpha-1}$$

$$M''(\theta) = \alpha\beta^2(\alpha + 1)(1 - \beta\theta)^{-\alpha-2}$$

$$M'''(\theta) = \alpha\beta^3(\alpha + 1)(\alpha + 2)(1 - \beta\theta)^{-\alpha-3}$$

$$\begin{aligned}
E(X^3) &= (\alpha^2\beta^3 + \alpha\beta^3)(\alpha + 2) = \alpha^3\beta^3 + \alpha^3\beta^3 + 2\alpha\beta^3 + 2\alpha\beta^3 = \alpha^3\beta^3 + 3\alpha^2\beta^3 + 2\alpha\beta^3 \\
&= \alpha^3\beta + 3\alpha^2\beta^3 + 2\alpha\beta^3 - \alpha^2\beta^3 - \alpha^3\beta \\
&= 2\alpha^2\beta^3 + 2\alpha\beta^3
\end{aligned}$$

Exercise 16.

- N people enter an elevator at the ground floor
- N — *poisson* distribution with parameter λ
- No one enters above the ground floor
- There are X floors above the ground floor, selected uniformly at random

$$P(\text{no one selects floor } i \mid N = n) = \left(1 - \frac{1}{k}\right)^n$$

$$\begin{aligned}
P(\text{no one selects floor } i) &= \sum_{n=0}^{\infty} \frac{\lambda^n e^{-\lambda}}{n!} \times \left(1 - \frac{1}{k}\right)^n \\
&= e^{-\lambda} \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} \left(1 - \frac{1}{k}\right)^n
\end{aligned}$$

Using Taylor Series for e^x

$$\begin{aligned}
&= e^{-\lambda} e^{\lambda(1-1/k)} \\
&= e^{-\lambda} e^{\lambda - \lambda \frac{1}{k}} \\
&= e^{-\lambda \frac{1}{k}}
\end{aligned}$$

$$P(\text{elevator stops at floor } i) = 1 - e^{-\lambda \frac{1}{k}}$$

$$\begin{aligned}
E(\text{number of stops elevator makes}) &= \sum_{i=1}^k (1 - e^{-\lambda \frac{1}{k}}) \\
&= k(1 - e^{-\lambda \frac{1}{k}})
\end{aligned}$$